

Compact ReusePlan record: 32-bit budget and edge substitution

The compact record substitutes for the structural edge word; an unsupported width fails closed.

COMPACT RECORD FIELDS AND 32-BIT BUDGET

destination	tier	epoch1	epoch2
16 bits	2 bits	5 bits	5 bits

$16 + 2 + 5 + 5 = 28$ bits: 65,536 vertices, 32 epochs

destination	epoch1	epoch2
18 bits	7 bits	7 bits

$18 + 0 + 7 + 7 = 32$ bits: 262,144 vertices, 128 epochs

$18 + 2 + 7 + 7 = 34$ bits: rejected, no decoder implements it

Substitution for the structural edge word

4-byte compact record replaces the 4-byte CSR edge word

8-byte general record instead adds a word per edge

weighted SSSP: 64-bit substitute, or edge plus 32-bit sidecar

Field meaning

destination: property vertex

tier: optional carried hint

epoch1, epoch2: two future

epochs for the same block

unused high bits stay zero

bound record width

Fail-closed format and sidecar binding

format word carries id_bits,

epoch_bits, tier_bits, presence

only tier_bits 0 or 2 decode

sidecar v2 binds the tier width

width mismatch aborts the run